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Unit – 5 : Session – 3 : SLO – 2 | Simulating Pattern Recognition with Input Strings

■ I. Questions & Answers

1. DFA for "ends in 01" — accepts exactly the strings over $\{0,1\}$ whose last two symbols are 01, e.g. 01, 001, 101, 1101, 10101. It rejects ϵ , 0, 1, 00, 10, 11, 100, etc.
2. Simulating input '1010' (DFA ends-in-01, states q_0 =start/no-match, q_1 =last symbol 0, q_2 =accept/ends-01):
read 1 $\rightarrow q_0$; read 0 $\rightarrow q_1$; read 1 $\rightarrow q_2$ (accept state); read 0 $\rightarrow q_1$. String exhausted at q_1 , which is not accepting $\rightarrow$ '1010' is REJECTED (it ends in 10, not 01).
3. Testing pattern recognition in code — with automata-lib, run `dfa.read_input_stepwise("1010")` to trace the state at each symbol, or `dfa.accepts_input("1010")` for the verdict; a hand-written simulator loops: $state = \delta(state, symbol)$ for each character, then checks $state \in F$.